

Ethan Harvey

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Highlights

- Fifth year Ph.D. candidate in the Machine Learning Research Group at Tufts University
 - Presented papers at NeurIPS (Journal-to-Conference Track and Workshop) and received an AISTATS Spotlight presentation (top 2.5% of submissions)
 - 5+ years of academic experience with implementation details of deep neural networks using machine learning frameworks including PyTorch, TensorFlow, and JAX
 - 2+ years of teaching assistant experience for the graduate level machine learning course (Python) and data structures (C/C++) at Tufts University
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Education

Tufts University, Medford, MA

Fall 2021 – Expected May 2026

Ph.D. in Computer Science, Cumulative GPA: 4.0

Advisor: Michael C. Hughes

Dissertation Title: *Bayesian Transfer Learning from Limited Labeled Data*

Relevant Coursework: Convex Optimization, Learning from Limited Labeled Data, Generative Models, Deep Neural Networks, Statistical Pattern Recognition, Machine Learning, Mathematical Aspects of Data Analysis

Tufts University, Medford, MA

Fall 2021 – Spring 2023

M.S. in Computer Science, Cumulative GPA: 4.0

Grove City College, Grove City, PA

Fall 2017 – Spring 2021

B.S. in Computer Science, Mathematics minor

Experience

Tufts University, Medford, MA

September 2021 – Present

A private R1 university.

Research Assistant, Advisor: Michael C. Hughes

- Collaborating with Tufts Medical Center and Kaiser Permanente Southern California to develop deep neural networks to classify covert cerebrovascular disease and predict time-to-event outcomes for stroke and dementia using CT and MRI brain scans.
- Developed a probabilistic method to extrapolate classifier accuracy to larger datasets that led to key figure on data adequacy in NIH grant R01-NS134859 (funded \$3 million).
- Priority user of Tufts University high performance computing (HPC) system including 40 NVIDIA A100 GPUs.

Philips, Cambridge, MA

June 2025 – August 2025

Medical Imaging Intern, Image Guided Therapy Devices R&D

- Benchmarked semi-supervised learning methods for mitral valve detection and segmentation in echocardiogram videos.

Philips, Cambridge, MA

May 2022 – August 2022

Machine Learning Intern, Philips Research North America

- Developed an ensemble tree model from millions of patient records for early detection of hospital-acquired infections.

- Compared general and infection-specific machine learning models.

UnitedHealth Group, Minnetonka, MN

June 2021 – August 2021

Data Science Intern, UnitedHealth Group R&D

- Developed a deep neural network to predict blood glucose values using sleep metrics from wearable fitness data and blood glucose monitoring data.
- Worked with internal data scientists on reinforcement learning project using multi-armed bandit algorithms.

Grove City College, Grove City, PA

January 2021 – May 2021

An ABET accredited college.

Research Assistant, Advisor: Jonathan O. Hutchins

- Collaborated with Highmark Health to research the application of unsupervised learning on acoustic signals as a surrogate for quantitative measurements of human foot pressures during walking and running.

Papers

- [7] **Ethan Harvey***, Dennis Johan Loevlie*, Amir Ali Satani, Wansu Chen, David M. Kent, and Michael C. Hughes. A Multi-Dataset Benchmark of Multiple Instance Learning for 3D Neuroimage Classification. In *Conference on Health, Inference, and Learning (CHIL)*, 2026.
- [6] **Ethan Harvey** and Michael C. Hughes. Occam’s Razor is Only as Sharp as Your ELBO. In *AISTATS 2026 Workshop on Optimisation and Post-Bayesian Inference in Machine Learning*, 2026.
- [5] **Ethan Harvey**, Mikhail Petrov, and Michael C. Hughes. Learning Hyperparameters via a Data-Emphasized Variational Objective. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026. (Spotlight presentation, top 2.5% of submissions).
- [4] **Ethan Harvey**, Dennis Johan Loevlie, and Michael C. Hughes. Synthetic Data Reveals Generalization Gaps in Correlated Multiple Instance Learning. In *Machine Learning for Health (ML4H) Findings Track*, 2025.
- [3] **Ethan Harvey**, Mikhail Petrov, and Michael C. Hughes. Learning the Regularization Strength for Deep Fine-Tuning via a Data-Emphasized Variational Objective. In *NeurIPS 2024 Workshop on Fine-Tuning in Modern Machine Learning: Principles and Scalability*, 2024.
- [2] **Ethan Harvey***, Mikhail Petrov*, and Michael C. Hughes. Transfer Learning with Informative Priors: Simple Baselines Better than Previously Reported. *Transactions on Machine Learning Research (TMLR)*, 2024. (Presented at NeurIPS 2024 poster session in Journal-to-Conference Track).
- [1] **Ethan Harvey**, Wansu Chen, David M. Kent, and Michael C. Hughes. A Probabilistic Method to Predict Classifier Accuracy on Larger Datasets given Small Pilot Data. In *Machine Learning for Health (ML4H)*, 2023. (28.7% acceptance rate).

Technical Skills

Programming Languages: Python, C/C++, Java, MATLAB

Machine Learning Frameworks: PyTorch, TensorFlow, JAX, scikit-learn

Data Analysis Tools: NumPy, Pandas, SciPy, Matplotlib, Spark, SQL, NoSQL

Cloud Computing Services: Amazon Web Services, Google Cloud Platform, Microsoft Azure

Teaching

CS 135 Machine Learning Teaching Assistant, Tufts University, Medford, MA

Spring 2022, Fall 2022, Spring 2023

- Developed Python unit testing framework that automatically grades student assignments.
- Led team of eight graduate and undergraduate teaching assistants to hold office hours and grade assignments.

- Taught weekly recitations on topics including linear and polynomial regression, perceptron learning, evaluating model performance, logistic regression, kernels, support vector machines, neural networks, convergence and non-convergence, parameter tuning, decision trees, dynamic programming, reinforcement learning, and k-means clustering.

CS 15 Data Structures Teaching Assistant, Tufts University, Medford, MA

Fall 2021

- Lectured on topics including lists, complexity, queues, stacks, recursion, binary search, sets, and trees.
- Worked on infrastructure team to design, maintain, and test programs that automatically grade student assignments.

Awards & Scholarships

Tufts University School of Engineering 2026 Outstanding Academic Scholarship Award

Award based on academic merit.

Tufts University Full Tuition Scholarship, 2021, 2022, 2023, 2024, 2025

Approximately \$10,000 yearly scholarship based on academic merit.

Dean's List with High Distinction, Grove City College, Spring 2021

Honor based on academic merit.

Grove City College General College Scholarship 2020

\$4,000 scholarship based on academic merit.

Grove City College Arlene and James Adams Scholarship 2019

\$3,400 scholarship based on academic merit.

Grove City College Timson-Eaton Student Aid Fund 2018

\$4,100 scholarship based on academic merit.

Academic Services

Area Chair:

- 2026 – Machine Learning for Healthcare (MLHC)

Reviewer:

- 2026 – International Conference on Artificial Intelligence and Statistics (AISTATS); AISTATS Workshop on Optimisation and Post-Bayesian Inference in Machine Learning (OPTIMAL@AISTATS); Conference on Health, Inference, and Learning (CHIL); Advances in Neural Information Processing Systems (NeurIPS)
- 2025 – Conference on Health, Inference, and Learning (CHIL); Machine Learning for Healthcare (MLHC); Machine Learning for Health (ML4H); NeurIPS Workshop on GenAI for Health: Potential, Trust, and Policy Compliance (GenAI4Health@NeurIPS)
- 2024 – Conference on Health, Inference, and Learning (CHIL); Machine Learning for Healthcare (MLHC); Machine Learning for Health (ML4H); NeurIPS Workshop on Fine-Tuning in Modern Machine Learning: Principles and Scalability (FITML@NeurIPS)